# ESTIMATING ENGEL CURVES: EVIDENCE FROM INDONESIAN HOUSEHOLD DATA




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Submitted by
**Muhammad Hanif Rizki**
**12020118190202**

**FACULTY OF ECONOMICS AND BUSINESS**
**DIPONEGORO UNIVERSITY**
**SEMARANG**
**2022**

## SUPERVISOR APPROVAL

| | | |
|---|---|---|
| Name | : | Muhammad Hanif Rizki |
| Student ID. | : | 12020118190202 |
| Faculty /Program | : | Faculty of Economics and Business /Economics |
| Thesis Title | : | Estimating Engel Curves: Evidence from Indonesian Household Data |
| Supervisor | : | Dr.rer.pol. Alfa Farah, S.E., M.Sc. |

Semarang, July 21, 2022

Thesis Supervisor



Dr.rer.pol. Alfa Farah, S.E., M.Sc.

NIP. 19830405 200912 2 008

## ORAL EXAM REPORT

| | | |
|---|---|---|
| Name | : | Muhammad Hanif Rizki |
| Student ID. | : | 12020118190202 |
| Faculty / Program | : | Faculty of Economics and Business/ Economics |
| Thesis Title | : | Estimating Engel Curves: Evidence from Indonesian Household Data |
| Supervisor | : | Dr.rer.pol. Alfa Farah, S.E., M.Sc. |

**has passed an oral examination on Thursday, 28 July 2022**

Examining Committee

1. Dr.rer.pol. Alfa Farah, S.E., M.Sc. .....
2. Prof. Dr. FX. Sugiyanto, M.S. .....
3. Wahyu Widodo, SE., M.Si., Ph.D. .....

Mengetahui,

Wakil Dekan Bidang Akademik dan Kemahasiswaan

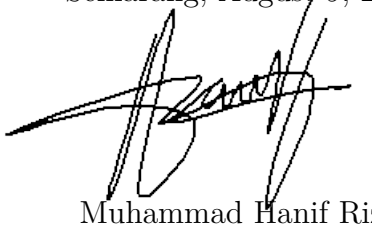
Firmansyah, S.E., M.Si, Ph.D

NIP. 19740427 199903 1 001

## STATEMENT OF ORIGINALITY

I, Muhammad Hanif Rizki, hereby declare that this thesis titled "Estimating Engel Curves: Evidence form Indonesian Household Data" is solely my own work. I have clearly referenced, in both the text and the bibliography or references, all sources used in the work. I understand that any false claim and academic misconduct in respect of this work will result in disciplinary action in accordance with university regulations, including revocation to the degree.

Semarang, August 9, 2022



Muhammad Hanif Rizki

12020118190202

"*Ask five economists and you'll get five different answers - six if one went to Harvard.*"

Edgar Fiedler

*For my parents!*

## ABSTRACT

Understanding consumer behavior is central to economics. This study empirically estimates the Engel curve for Indonesian households and investigates Indonesian households' consumption patterns for selected commodities. Using the IFLS data wave 5, covering 15,118 households across Indonesia and various Engel curve specifications, the study found that food and fuel are classified as necessities, whereas clothing, transportation, and other goods are luxury goods. Food, transportation, and other goods are non-linear, while clothing and fuel are linear in parameters. Furthermore, this study used instrumental variables to moderate the endogeneity. The results of different Engel curve specifications show significance in Indonesian household data with the existence of apparent heteroskedastic errors due to heterogeneity in preferences. In general, the household consumption pattern is shown to be consistent with the Engel law and the previous evidence.



**Keywords:** Engel Curves, Household Consumption, Engel Law

**JEL classification: D12, D11, D00**

## *ABSTRAKSI*

*Memahami perilaku konsumen merupakan inti dari studi ekonomi. Penelitian ini secara empiris memperkirakan kurva Engel untuk rumah tangga Indonesia dan menyelidiki pola konsumsi rumah tangga Indonesia untuk komoditas tertentu. Dengan menggunakan data IFLS gelombang 5 yang mencakup 15.118 rumah tangga di Indonesia dan berbagai spesifikasi kurva Engel, penelitian ini menemukan bahwa pangan dan bahan bakar rumah tangga tergolong necessity, sedangkan sandang, transportasi, dan barang lainnya tergolong luxury. Makanan, transportasti, dan barang lainnya bersifat non-linier, sedangkan sandang dan bahan bakar rumah tangga bersifat linier dalam parameter. Selain itu, penelitian ini menggunakan variabel instrumental untuk memoderasi endogenitas. Hasil spesifikasi-spefikasi kurva Engel menunjukkan signifikansi pada data rumah tangga Indonesia dengan adanya heteroskedastisitas yang tampak karena heterogenitas preferensi. Secara umum, pola konsumsi rumah tangga terbukti konsisten dengan hukum Engel dan penelitian sebelumnya.*



**Kata kunci:** Kurva Engel, Konsumsi Rumah Tangga, Hukum Engel

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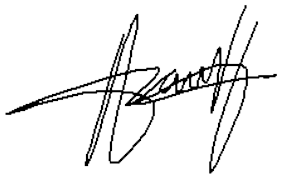
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The author realizes the flaws and errors in the preparation, presentation, and writing of this thesis. The author hopes for constructive criticism and suggestions for the improvement of this thesis. The author hopes that this thesis can add insight and knowledge to anyone who reads it.

Semarang, 19 July 2022



Muhammad Hanif Rizki

12020118190202

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# CHAPTER I

# INTRODUCTION

## 1.1 Background

Understanding how consumers behave is central to the study of economics. To understand and capture how consumers behave, economic models rely on two fundamental assumptions, namely methodological individualism and rationality. A rational consumer allocates his or her limited resources to maximize his or her utility. The Engel curve particularly studies how consumers allocate their varying budget or income to a commodity or a service.

In general, Engel curves are demand curves that depict the relationship between individual or household spending on certain commodities and total income or expenditure (Lewbel, 2008). Engel curves are often used to understand household consumption patterns and to find a commodity's income elasticity of demand. The income elasticity of demand is used for classifying commodities as either inferior or normal.

The Engel curve was originated from a German economist Ernst Engel (Zimmerman, 1932). The first development in estimating empirical Engel curve was due to Allen and Bowley (1935). It was then popularized by Working (1943) and Leser (1963). Estimating Engel curves has always been concerned with the correct specifications. Some studies compares different types of specifications ranging from parametric to non-parametric estimation, for example Leser (1963) and Engel and Kneip (1996). Furthermore, estimating Engel curves has been shown to have complications such as heteroskedasticity, functional forms, and

endogeneity.

Many studies has empirically estimate Engel curve. Most of them employed data from developed countries. For example: Working (1943) for the U.S.; Leser (1963) for the U.S. and Ireland; Liviatan (1961) for Britain and Israel; Giles and Hampton (1985) for New Zealand; Banks et al. (1997) and Blundell, Duncan, and Pendakur (1998) for the U.K.; and Battistin and Nadai (2015) for Italy. However, only few employed data from developing countries. For example: Deaton (1985) for Sri Lanka; De Vreyer et al. (2020) for Senegal; and Suharno (2010) for Indonesia.

In general, the findings suggests that food is considered as a necessity and is consistent with the Engel Law, while clothing and other goods are considered as luxury goods.

The aim of this study is to empirically estimate the Engel curve using Indonesian household data. Currently classified as a developing and middle-income country, Indonesia is the fourth most populated country and one of the world's fastest growing consumer markets. Given Indonesia's cultural and demographic diversity, consumer patterns might vary given their income.

In addition, this study estimates various Engel curve specifications in order to better understand household budget allocation and to calculate the income elasticity of demand for selected commodities.

## 1.2 Research Questions

This study addresses the following questions: how does Indonesian households consumption for selected commodities varies with household income?

## 1.3 Research Aims and Objectives

The aim of the study is to investigate Indonesian households' consumption for selected commodities. In order to achieve the aim of the study, the objectives of the study are:

1. to empirically estimate Engel curves for Indonesian households
2. to calculate income elasticities of the selected commodities

## 1.4 Thesis Outline

This thesis is structured into 5 chapters. Chapter 1 describes the background and motivations for estimating empirical Engel curves. Chapter 2 describes the literature behind Engel curves from a theoretical and empirical standpoint. It discusses the empirical development of models and its findings in estimating empirical Engel curves. Chapter 3 describes the research method, which includes the data explanation, operational variables definitions, data analysis method, model specifications, and statistical inference. Chapter 4 describes the descriptive statistics and explores in detail the Indonesian household characteristics. The chapter further discusses the estimation results, the comparison of various specifications, and the empirical evidence of Engel curves. Chapter 5 describes the conclusions of the research results. In addition, it includes research limitations and suggestions for further research.

# CHAPTER II

# LITERATURE REVIEW

## 2.1 Theoretical Literature

### 2.1.1 Engel Curve Definition and Its Historical Context

Engel curves depict the relationship between individual or household spending on a certain commodity or service and total income or expenditure (Figure 2.1). Engel curves are frequently used to estimate the income elasticity of demand and, hence, to classify whether or not a good is an inferior or a normal good. Normal goods are further classified as necessity and luxurious. Generally, a negative slope indicates that the commodity is considered an inferior good, while a positive slope indicates a normal good. The slope of the curve may often be linear or highly non-linear for different types of goods.

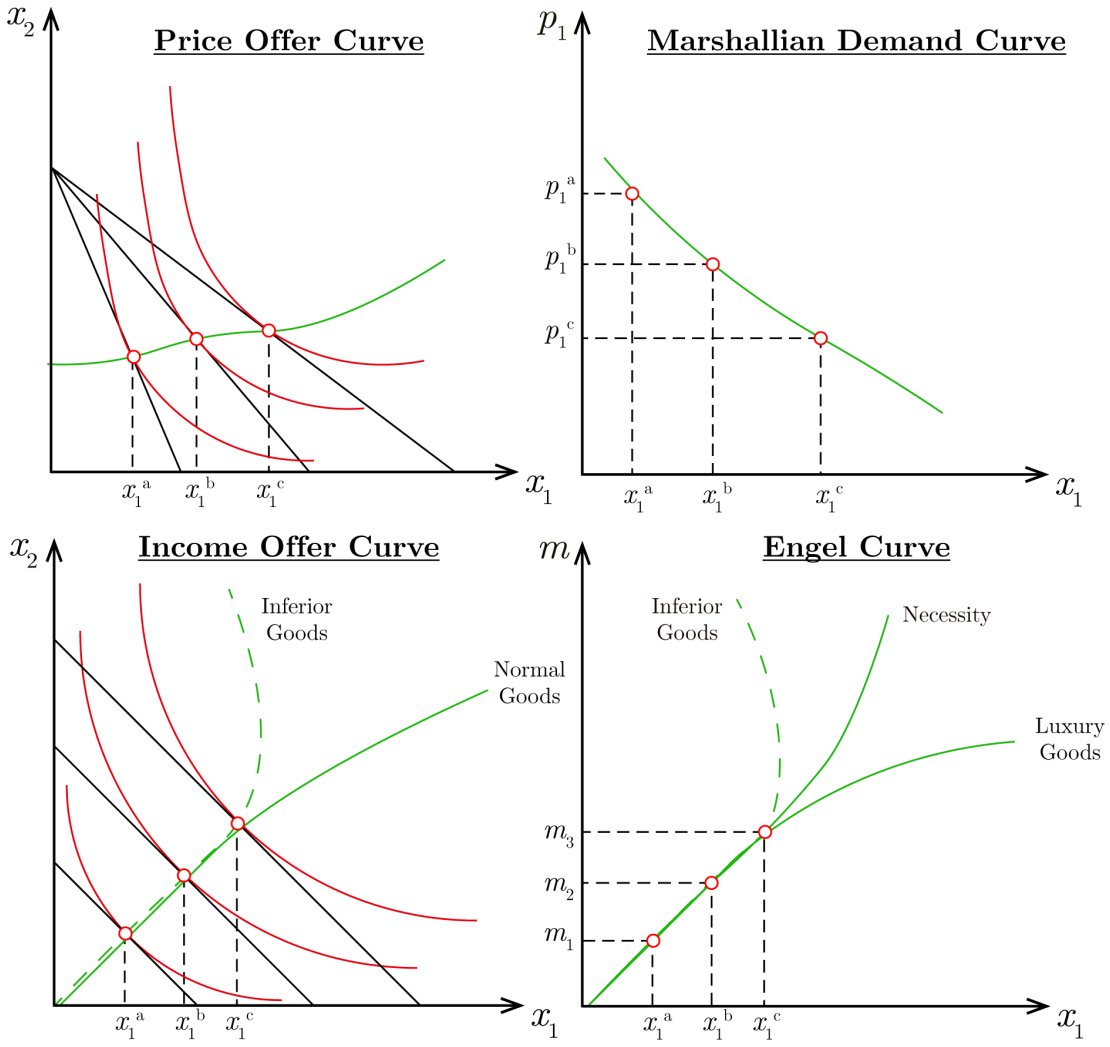
An Engel curve is derived from a Marshallian demand function. For good $i$ consumed by household $h$, the Marshallian demand function is expressed as:

$$x_{ih} = x_{ih}(m_h, \bar{z_h}, \bar{p_i}, \ldots, \bar{p_j}), \tag{2.1}$$

where $x_{ih}$ is defined as the quantity of good $i$ consumed by household $h$. It is a function of total household expenditure or income $m_h$, household characteristics $z_h$, price of the good $p_i$ and other prices $p_j$. In the Engel curve specification, income changes while the other factors do not.

The Engel curve originated and is named after a German economist and statistician named Ernst Engel. Engel wrote his famous work in 1857 entitled "The

FIGURE 2.1: Engel Curves




Consumption−Production Relations in the Kingdom of Saxony" (Chai and Moneta, 2010). In this article, Engel debated the Malthusian hypothesis, which is a hypothesis proposed by Malthus et al. (1992) in which the growth of the population increases geometrically while food production increases arithmetically. Since the population outpaces food production, shortages and crises arise; this event is commonly referred to as the Malthusian catastrophe. Engel argued that the catastrophe is avoidable as long as consumer demand and consumption patterns adapt to the growing economy's production capacity.

To support his argument, Engel investigated how consumption patterns change as income changes. In his investigation, Engel found that there exists a pattern in the relationship between commodities consumed and households' income. The pattern is then famously known as the Engel law. The Engel law states that as household income increases, the percentage spent on food decreases (Zimmerman, 1932). Additionally, Engel found that the proportions of expenditure on clothing, rent, and fuel stay the same, while that for sundries increases. The pattern implies that households with lower incomes must fulfill their basic needs first (food), and then, as they get wealthier, their income is spent on other goods.

While examining household consumption patterns, Engel concluded that the expenditure patterns reflect consumer welfare in a sense that how goods can be classified as a "necessity" or a "luxury" (Chai and Moneta, 2010). Engel further concluded that food, as a basic need, was a priority to fulfill at the lower levels of income.

### 2.1.2 Models in the Parametric Estimation of Engel Curve

For empirical analysis, Engel curves must satisfy several consumer theory assumptions and demand function properties. Households are assumed to choose the number of bundles of goods that maximize their utility level, subject to their budget constraint. Furthermore, Engel curves assume that all prices across goods are constant and that households face the same market price at a given time and location.

Allen and Bowley (1935) developed the earliest empirical models for estimating the Engle curve. The authors introduced a model in which it is derived directly from the Marshallian demand function as in Equation (2.1). The specification is

as follows:

$$gi = \alpha_i + \beta_i y + \varepsilon_i \tag{2.2}$$

where $g_i$ is defined as the expenditure for good $i$, $y$ is the total income of the household, $\alpha_i$ and $\beta_i$ are parameters, and $\varepsilon_i$ is the error term. The specification is then found to have large errors and does not fit very well (Leser, 1963).

Working (1943) then modified the specification. Instead of using the expenditure of goods consumed $g_i$ and income $y$, Working (1943) used the share of expenditure spent on good $i$ ($w_i$) and total expenditure $m$. Here, $w_i = \frac{g_i}{m}$. One reason for using total expenditure in contrast to using income is to eliminate the problem of allocation of income. Some households have a high propensity to save, while others do not. This makes the variation across households large, hence the creating difficulties in estimating and predicting the Engel curves.

The specification was later tested by Leser (1963) on household survey data from the U.S. and Ireland. The author found that the specification fit pretty well and was consistent across goods. This model is famously known as the Working−Leser specification and is frequently used to estimate empirical Engel curves. The specification is as follows:

$$w_i = \alpha_i + \beta_i \log m + \varepsilon_i \tag{2.3}$$

The logarithmic function ensures the viability of estimating Engel curves since households' total expenditure is extremely variable and also gives a more useful interpretation as opposed to use in the nominal form.

This form is known as a log-linear budget-share demand system in which it satisfies several demand properties.[1] The specification has a restriction in which the sum of budget shares must equal to one. This restriction is known as the *adding-up* property.

$$\sum_{i=1}^{N} w_i = 1 \; ; \sum_{i=1}^{N} \alpha_i = 1 \; ; \sum_{i=1}^{N} \beta_i = 0 \tag{2.4}$$

Furthermore, in this form, the income elasticity of demand can be calculated. To calculate the income share elasticity of demand we take the partial derivative with respect to $\log(m)$. The income elasticity of demand for good $i$ is the percent change in $w_i$ that results from a percent change in $\log m$ plus one.

$$\eta_i = \frac{\partial \log(\alpha_i + \beta_i \log m)}{\partial \log m} + 1, \tag{2.5}$$

where the income elasticity of demand can be re-written as (Nicholson and Snyder, 2012)

$$\eta_i = \eta_{w_i} + 1, \tag{2.6}$$

where $\eta_i$ denotes for the income elasticity of demand and $\eta_{w_i}$ denotes for the income share elasticity of demand.

Goods that have an income elasticity of less than zero are considered inferior, whereas goods with an income elasticity of greater than zero are considered normal goods. Normal goods are then classified as necessities and luxuries. [2]

[1] The properties of the Engel curve demand system are aggregability and rank (Lewbel, 2008)

[2] Necessity goods refer to goods that consumers purchase regardless of their income levels. An increase in income will generally not increase the amount or proportion spent on that good. If the income elasticity is $0 < \eta_i < 1$, it is a necessity. As for luxury goods, as a consumer's income rises, their proportion of consumption of those goods will tend to increase as well. If the income elasticity is $\eta_i > 1$, it is a luxury good.

With the *adding-up* property and the income elasticity of demand, one condition that empirical Engel curves estimation must meet is the Engel Aggregation (Deaton and Muellbauer, 1980b).

The Engel Aggregation states that the sum of all expenditure share of goods $i$ times its income elasticity is equal to one (The share-weighted income of elasticity is equal 1). This condition is met as consumers are assumed to exhaustively spend all their expenditure. (See Appendix A).

$$\sum_{i=1}^{N} w_i \eta_i = 1 \tag{2.7}$$

Deaton (1985) modified the Working-Leser specification for Sri Lankan household data by including the quadratic term to better fit the model and controls to reduce the variation in the error.

$$w_i = \alpha_i + \beta_{1i} \log m + \beta_{2i} \log m^2 + \sum_{k=1}^{n} \gamma_k z_k + \varepsilon_i, \tag{2.8}$$

where $\gamma_k$ are the parameters for controls $z_k$.

The Working-Leser specification motivates the development of the Almost Ideal Demand System (AIDS, Deaton and Muellbauer, 1980a) and its extension, Quadratic Almost Ideal Demand System (QUAIDS, Banks et al., 1997). The demand systems and, thus, the derived Engel curve, are widely adopted because they allow estimation of both price and income elasticities across goods. The specification satisfies the additive property, the Slutsky symmetry, and is homogenous to a degree of zero in prices and total expenditure. However, this specification requires that the prices of goods be observed. The following is the AIDS model, where $\beta_i$

is the parameter for the Engel curve.

$$w_i = \alpha_i + \sum_{j=1}^{n} \gamma_{ij} \log p_j + \beta_i \log(\frac{m}{P}) + \sum_{k=1}^{n} \delta_k z_k + \varepsilon_i \tag{2.9}$$

### 2.1.3 Problems in the Parametric Estimation of Engel Curve

Several complications commonly arise when estimating Engel curves. The complications include: (1) heteroskedasticity, (2) functional forms and the shape of the Engel curve, and (3) endogeneity.

Heteroskedastic errors are probably the most prominent complication with which most empirical studies deal. Heteroscedastic errors arise due to unobserved heterogeneities, such as tastes, the quality of goods purchased, and the different prices that consumers face (Allen and Bowley, 1935). This leads to inconsistent estimates and low explanatory power (low R-squared). However, heteroskedastic errors can be minimized by including control variables in the model, such as measurable household characteristics, demographics, prices of all goods being observed, and other controls that relate to demand (Lewbel, 2008). Additionally, white robust standard errors allow for valid statistical inferences in the presence of heteroscedastic errors (White, 1980).

Specifying the functional form and, thus, defining the shape of Engel curves has also been an issue when estimating Engel curves. Misspecification in the empirical Engel curve model lead to inconsistent estimates (Engel and Kneip, 1996). The misspecification that lead inconsistency is due to the nature of the type of good. Goods can sometimes be linear or non-linear in parameters (Working, 1943). The non-linearity could be due to consumers' proclivity to purchase goods of

varying quality based on their income level (Suharno, 2010). In other words, as consumers income increases, consumers tend to purchase higher-quality goods. This, in turn, suggests that the income elasticity of demand for goods changes as income increases. This implies that some goods are luxury to lower levels of income and necessity at a higher level (Banks et al., 1997).

Other empirical work, such as Banks et al. (1997), Blundell, Duncan, and Pendakur (1998), and Liviatan (1961) discovered that endogeneity has a significant impact on estimating the curvature of the Engel curve. Because of the endogeneity problem, the estimation is biased. The authors found that using earned or disposable income as an instrumental variable helped moderate the bias. Battistin and Nadai (2015) suggests mean wages as an ideal instrument.

## 2.2 Empirical Evidence

Empirical evidence on the relation of proportion spent on food, such as Banks et al. (1997), Blundell, Duncan, and Pendakur (1998), De Vreyer et al. (2020), Deaton (1985), Giles and Hampton (1985), Leser (1963), and Working (1943), suggests that the Engel law is consistent for household data regardless of geographic location, occupancy of household head, age of household head, and household size. Generally, the relationship between the proportion spent on food and log total expenditure is linear. Furthermore, food is considered a necessity and has a significant effect on household size; that is, larger families tend to spend more on food.

For other commodities, the proportion spent on clothing and other goods increases as income increases, and is considered a luxury good as estimates on the income elasticity of demand are larger than unity (Blundell, Duncan, and

Pendakur, 1998; De Vreyer et al., 2020; Giles and Hampton, 1985; Leser, 1963; Working, 1943). However, the shape of the Engel curve is less consistent across communities studied, for example, Blundell, Duncan, and Pendakur (1998) found a linear relationship in clothing. This might suggest differences in heterogeneity in taste and differences in the quality of goods across communities.

Suharno (2010) estimated Engel curves from the almost ideal demand system for several food categories employed from the Indonesia dataset SUSENAS. It is found that rice, meat, and edible oil are income inelastic (necessity) for every household characteristic, while fish, fruits, and vegetables exhibit mixed income elasticities, and tobacco and prepared food are luxurious.

# CHAPTER III

# RESEARCH METHODS

## 3.1 Variables and Definitions

### 3.1.1 Dependent Variable

The dependent variable is the budget share on commodity $i$, denoted as $w_i = \frac{g_i}{x}$, where $g_i$ denotes weekly household expenditure on commodity $i$ and $x$ is the weekly total expenditure of all commodities expended by the household both measured in Rupiah. It is important to note that $w_i$ is measured in percentage form. For example, a $w_i$ value of 0.80, where commodity $i$ is food, means that 80% of household's weekly total expenditure is spent on food.

Motivated by the works of Banks et al. (1997), Blundell, Duncan, and Pendakur (1998), Leser (1963), and Working (1943), the estimations include the following commodities: (1) food; (2) clothing; (3) fuel; (4) transportation; and (5) other goods. Table 3.1 shows the commodity description.

### 3.1.2 Explanatory Variable

The explanatory variable is a household's weekly total expenditure measured in Rupiah. A household's weekly total expenditure is the sum of its weekly expenses on food, clothing, transportation, fuel, and all other goods. It is important to note that in this variable, received goods that households consume are not counted as expenditure.

TABLE 3.1: Commodities Description

| Commodity $i$ | Description |
|---|---|
| Food | Staple foods; vegetables; dried foods; meats and fish; other dishes; milk/eggs; beverages; and other consumer products |
| Clothing | Shoes; hats; shirts; pants; children clothing; etc. |
| Transportation | Gasoline; bus fare; cab fare; vehicle repair costs; etc. |
| Fuel | LPG; kerosene; etc. |
| Other Goods | Electricity; water; telephone; household items and supplies; recreation and entertainment; medicinal costs; etc. |

In the estimation, expenditure is transformed into logarithmic form. The logarithmic form allows for percentage interpretation. In addition, the transformation will reduce the extreme variations and, thus, moderate the heteroskedasticity and distribution of the errors to allow for consistent estimation.

### 3.1.3 Control Variables

In order to minimize the potential omitted variable bias and reduce heteroskedastic errors, we include control variables that are not only intuitive and theoretical, but also have been proven significant from previous empirical studies on empirical Engel curves (Lewbel, 2008). The control variables are as follows:

1. Urban (Dummy)

   Urban is measured in terms of a dummy variable, where the value 1 is for a household in an urban area, and 0 for one in a rural area. Households in urban areas are expected to have less proportion of their expenditure on basic needs like food and clothing (Working, 1943). One reason is because households in the urban area might be wealthier than those in the rural

Table 3.2: Years of Education Description

| Years of Schooling | Highest Level of Schooling Attended by Household Head |
|---|---|
| 0 | None / Not yet attended school / Others / Don't Know |
| 6 | Elementary School; Islamic Elementary School (Mardrasah Ibtidaiyah); *Kejar Paket A* |
| 9 | Junior High General and Vocational; Islamic Junior High; (Madrasah Tsanawiyah); *Kejar Paket B* |
| 12 | Senior High General and Vocational; Islamic Senior High; (Madrasah Aliyah); *Pesantren*; *Kejar Paket C* |
| 15 | College D1, D2, D3 |
| 16 | Undergraduate (S1); Open University |
| 18 | Masters (S2) |
| 21 | Doctorate (S3) |

area.

2. Household Head Education (Years)

   Household head education is the years spent by a household head obtaining education. Table 3.2 provides the standard years spent on a certain level of education. Household head education may influence consumption patterns because a more educated household head presumably allocates budget more effectively.

3. Household Marital Status (Dummy)

   Household marital status is measured as a dummy variable where married takes the value 1, and 0 otherwise. Married household heads might also affect the varying decision making in budget allocation.

4. Household Head Age (Years Old)

   Household head age might affect household budget composition. Intuitively, as the household head's age progresses, the composition of budget changes. Older household heads might spend more on luxury goods.

5. Household Head Sex (Dummy)

   The variable is measured as a dummy variable, where the male household

head takes the value 1, while the female takes the value 0. Gender differences in household heads may have an impact on budget allocation because of preferences. Male household heads spend more on food because men typically have a larger appetite than women. As a result, women spend more money on other items such as clothing and cosmetics.

6. Household Size (Persons)

   The more members a household has, the more efficient its use of housing facilities such as furniture and vehicles (Working, 1943). In other words, the proportion spent on shareable goods tends to be lower for households with more members.

## 3.2 Data Source

This study employed household data from the Indonesian Family Life Survey Wave 5 (IFLS 5). IFLS is an on-going longitudinal survey in Indonesia representing 83% of the Indonesian population. The dataset is available publicly at RAND website (Strauss et al., 2016).

The IFLS survey is divided into two types of questionnaires: household level and individual level. It aims to collect socio-economic data, which includes consumption, health, income, welfare, community opinions, housing, and the environment. The IFLS survey has its advantages; that is, data includes a low attrition rate of 6% and includes a wide range of questions. On the other hand, the data must be correctly tracked and merged before being used for analysis, and some questions are changed or added over the course of waves.

To date, there are already five IFLS waves. The first wave (IFLS 1) was conducted in 1993/1994. The wave covered over 30.000 individuals residing in 13 provinces.

The second wave (IFLS 2) was fielded in 1997/1998. The third wave (IFLS 3) was fielded in the year 2000. The fourth wave (IFLS 4) was fielded in 2007/2008. The final and current wave (IFLS 5) was fielded in 2014/2015. IFLS 5 covered over 80,000 individuals and 15,000 households residing in 24 provinces,.

This empirical study uses data from the IFLS 5 covering all Indonesian households within the survey. This study particularly uses the questionnaire from BOOK K, BOOK 1, BOOK 2, and BOOK 3A covering household characteristics, household consumption, household economy, and individual economy, respectively.

The data on expenditures recorded on food are on a weekly basis. Non-food items (clothing, household supplies, furniture, medicinal costs, education, etc.) are recorded as yearly expenses, while other expenses like fuel, electricity, transportation, etc. are recorded as monthly expenses. Therefore, this study transformed all the expenditure data into a weekly schedule.

When dealing with survey data, zero expenditure problems likely arise (Suharno, 2010). The problem arises when households do not consume the commodities being analysed. One example of a zero expenditure problem is when households do not purchase a commodity as frequently. Another example of a zero expenditure problem is when certain commodities require the household to own certain commodities, for example, gasoline, which requires the household to own a vehicle. Another most likely reason for this problem is that households tend to receive commodities rather than purchase them, making the household consumption composition seem unreasonable. The problem of zero expenditure will lead to heteroskedastic errors and endogeneity in our estimations.

## 3.3 Data Analysis

### 3.3.1 Baseline Model Specification

We specify a variety of models to estimate the Engel curve for the selected commodities. The baseline model is the Working-Leser specification, expressed as follows:

$$w_{ihd} = \alpha_i + \beta_i \log m_{hd} + \delta_d + \sum_{k=1}^{n} \gamma_k z_k + \varepsilon_{ihd}, \tag{3.1}$$

where $w_{ihd}$ denotes the share of expenditure on commodity $i$ for household $h$ located in district $d$, $m_{hd}$ is the total expenditure of household $h$ located in district $d$ and $z_k$ is a set of control variables, which include: urban dummy, household head education, household marital status, household head age, household head sex, and household size. Here, the parameter of interest is $\beta_i$.

In the estimation, we also include district *fixed-effects* ($\delta_d$), to control for time-invariant heterogeneities across districts. Households located in a district presumably faces similar price and share similar tastes, preferences, and cultures to those located in two or more different districts. By including district *fixed-effects*, we compare households within a district, not across a district.

In addition to the baseline model, we also estimate a quadratic model. Deaton (1985) argues that the quadratic model is the better fit model to evaluate the shape of the Engel curve. The quadratic model is as follows:

$$w_{ihd} = \alpha_i + \beta_{1i} \log m_{hd} + \beta_{2i} \log m_{hd}^2 + \delta_d + \sum_{k=1}^{n} \gamma_k z_k + \varepsilon_{ihd} \tag{3.2}$$

The quadratic function are used to capture the increasing or diminishing effect of the total total expenditure. Accordingly, we could calculate the turning point

at which the function is at its maximum or minimum (Wooldridge, 2015). The turning point is calculated by:

$$\log m^*_{hd} = \left| \frac{\hat{\beta_{1i}}}{2\hat{\beta}_{2i}} \right| \tag{3.3}$$

We estimate equation (3.1) and (3.2) using Ordinary Least Squares.

We further check our estimates whether or not they satisfy the Gauss-Markov assumptions. The Gauss-Markov assumptions are as follows: (1) linear in parameters; (2) random sampling; (3) no perfect collinearity, that is, there is no exact relationship between the independent variables; (4) zero conditional mean, where the error term does not correlate with the main explanatory variable, $E(u \mid m_{hd}) = 0$; and (5) homoscedasticity, where the error $u$ has constant variance, $\text{Var}(u \mid x_1, \ldots, x_i) = \sigma^2$.

The Engel curve specifications in this budget share form must satisfy the Engel aggregation, that is, the *adding-up* property, where $i$ denotes for the selected commodity.

$$\sum_{i=1}^{N} w_i = 1 \; ; \sum_{i=1}^{N} \alpha_i = 1 \; ; \sum_{i=1}^{N} \beta_i = 0 \tag{3.4}$$

These restrictions on the demand system, in turn, would allow us for calculating the income elasticity of demand for commodity $i$ following Deaton (1985) and Teklu, Johnson, et al. (1986). (see Appendix A).

$$\eta_i = \frac{\beta_i}{\bar{w}_i} + 1, \tag{3.5}$$

where $\eta_i$ denotes the income elasticity of demand for good $i$, $\dfrac{\beta_i}{\bar{w}_i}$ denotes the income share elasticity of demand for good $i$ which could be denoted as $\eta_{w_i}$.

### 3.3.2 Dealing with Endogeneity Problem

The problem of endogeneity is typically present in household budget data. There are at least 3 sources of endogeneity; (1) omitted variable bias; (2) measurement errors; and (3) simultaneity. In our case, the data on total expenditure is suspected to be endogenous primarily caused by measurement errors (Battistin and Nadai, 2015), thus violating the Gauss-Markov *zero-conditional mean* assumption, resulting in biased estimates.

Omitted variable bias occurs when relevant variables are not included in the model. In our case, omitted variable bias is minimized by including control variables.

Measurement errors occurs when observed data does not reflect the true empirical value. Measurement errors occurs presumably because of the zero expenditure problem (Suharno, 2010).

Simultaneity bias occurs when where the explanatory variable is jointly determined with the dependent variable. In our case, total expenditure is endogenous because it is likely to be determined by income (Blundell, Duncan, and Pendakur, 1998; Liviatan, 1961).

In order to moderate the endogeneity problem, we employ instrumental variables (IV) method. The use of IV in the context of Engel curves is widely applied in empirical works such as Banks et al. (1997), Battistin and Nadai (2015), Blundell, Chen, and Kristensen (2007), Blundell, Duncan, and Pendakur (1998), and Liviatan (1961). Motivated by Blundell, Duncan, and Pendakur (1998), the instrumental variable used for this study is earned income. Earned income is defined as the total weekly wage, salary, or income (including all the benefits) received by

all the members of that household, measured in Rupiah. The variable does not include other sources of income like business profits, return on assets, non-labor income, retirement funds, government scholarships, etc.

The IV model is estimated using the 2SLS (Two Stage Least Squares) estimator. The variable log earned income ($\log y$) act as the instrument for log total expenditure ($\log m$). We estimate log total expenditure with log earned income in the first stage and then use the estimated log total expenditure in our original model for the second stage. The IV structural equation is as follows:

$$\begin{aligned} \log m_{hd} &= \pi_{0i} + \pi_{1i} \log y_{hd} + \delta_d + \sum_{k=1}^{n} v_k z_k + u_{ihd} \\ w_{ihd} &= \beta_{0i} + \beta_{1i} \widehat{\log m_{hd}} + \delta_d + \sum_{k=1}^{n} \gamma_k z_k + \varepsilon_{ihd} \end{aligned} \tag{3.6}$$

The instrumental variable must satisfy the following assumptions (Wooldridge, 2015), namely the instrument $\log y$ is uncorrelated with the error term $u$, and, $\log y$ is partially correlated with the independent variable $\log m$,

$$Cov(\log y \mid u) = 0, \tag{3.7}$$

$$Cov(\log y \mid \log m) \neq 0 \tag{3.8}$$

This implies that our instrument log earned income $\log y$ affects the budget share on commodity $w_i$ only through log total expenditure $\log m$. The idea is that a household's total expenditure has some part of it explained by their earned income since households might be dissaving. This way, $\widehat{\log m}$ is an unbiased estimate for $\log m$. It is worth noting that the $R^2$ produced by the IV estimation will be smaller than that in the OLS estimation and consequently, the $R^2$ on the IV estimates becomes meaningless (Wooldridge, 2015).

There are two important tests to evaluate the chosen IV. The first test is the Weak Instruments test to test whether our instrument $\log y$ is a strong instrument for $\log m$. The null hypothesis states that the instrument has a weak correlation to the endogneous variable, while the alternative states that the instrument has a strong correlation to the endogneous variable.

The second test is the endogeneity test. Because using the 2SLS estimator is less efficient compared to using OLS if all the independent variables are exogenous (Wooldridge, 2015), we test for endogeneity in our $\log m$ using the Wu-Hausman test (Hausman, 1978). The test informs whether or not the IV estimates are as consistent as the OLS estimates. The null hypothesis of the test states that the 2SLS and OLS estimates are equally consistent, while the alternative hypothesis states that the 2SLS and OLS are different, and, therefore, IV estimates are more consistent. Rejection of the null implies that our independent variable $\log m$ is not endogenous and OLS is preferred.

## 3.4 Statistical Inference

The parameter of interest is $\beta_i$. The statistical hypotheses are as follows:

$$H_0 : \beta_i = 0$$

$$H_1 : \beta_i \neq 0$$

We compare the probability of $t$-statistics (p-value) with a chosen significant level. If the p value is lower than the chosen significant level, we can reject the null hypothesis. This implies that the explanatory variable is statistically significant at the chosen significant level.

We also compare the goodness of fit with the adjusted $R^2$. $R^2$ takes the value between 0 and 1. A 0.30 value in $R^2$ means that 30% of the variation in the model is explained by the explanatory variables. Further, we also test for simultaneous effects of our independent variables using the F-test. A significant F-statistic shows that our model is capable of explaining the dependent variable.

# CHAPTER IV

# RESULTS AND DISCUSSION

## 4.1 The Average Indonesian Household

Our estimation include only 15,118 households out of 15,921 households (94.9%) provided in the IFLS 5 data. The observation are omitted because of missing values.

FIGURE 4.1: Household Sample Distribution by Island

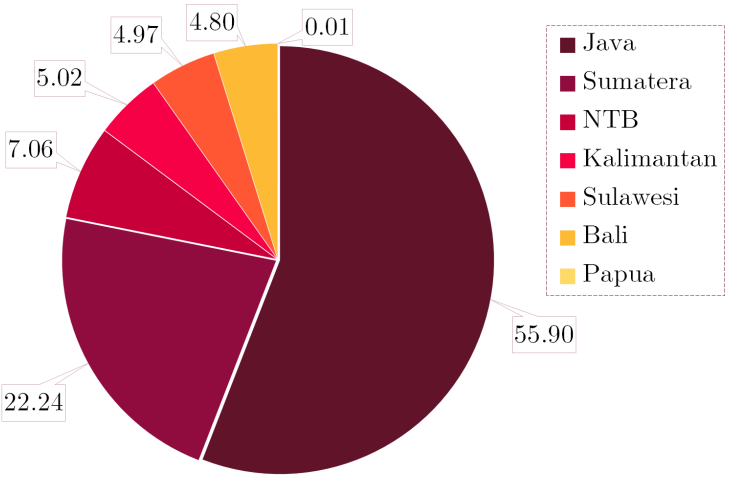



Figure 4.1 shows the household sample distribution by island. The sample is distributed unequally across islands, where most of the households are located in Java island, making up 55.9% of the total sample, followed by Sumatera island (22.24%), West Nusa Tenggara (7.06%), Kalimantan island (5,02%), Sulawesi island (4.97%), Bali island (4.80%), and Papua island (0,01%). However, this

TABLE 4.1: Observed Household Provinces

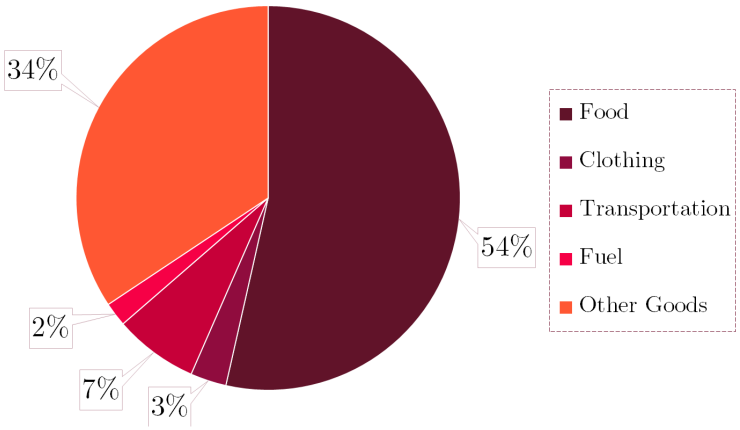
| No. | Province | n | Percent |
|---|---|---|---|
| 1 | Aceh | 2 | 0,01 |
| 2 | North Sumatera | 1141 | 7,55 |
| 3 | West Sumatera | 648 | 4,29 |
| 4 | Riau | 130 | 0,86 |
| 5 | Jambi | 21 | 0,14 |
| 6 | South Sumatera | 687 | 4,54 |
| 7 | Bengkulu | 603 | 3,99 |
| 8 | Lampung | 93 | 0,62 |
| 9 | Kep. Bangka Belitung | 37 | 0,24 |
| 10 | DKI Jakarta | 969 | 6,41 |
| 11 | West Java | 2147 | 14,20 |
| 12 | Central Java | 1859 | 12,30 |
| 13 | Yogyakarta | 804 | 5,32 |
| 14 | East Java | 2070 | 13,69 |
| 15 | Banten | 602 | 3,98 |
| 16 | Bali | 725 | 4,80 |
| 17 | West Nusa Tenggara | 1068 | 7,06 |
| 18 | West Kalimantan | 2 | 0,01 |
| 19 | Central Kalimantan | 25 | 0,17 |
| 20 | South Kalimantan | 670 | 4,43 |
| 21 | East Kalimantan | 62 | 0,41 |
| 22 | South Sulawesi | 723 | 4,78 |
| 23 | West Sulawesi | 29 | 0,19 |
| 24 | West Papua | 1 | 0,01 |
| | Total | 15,118 | 100 |

distribution is nearly identical with the distribution of population from the Indonesia's population census conducted by the Indonesian Statistics (BPS, Indonesia, n.d.). Therefore, the sample is adequate to represent the whole Indonesian population.

Looking at the distribution across provinces shown in Table 4.1, the majority of households are located in four provinces, namely West Java (14.20%), East Java (13.69%), Central Java (12.30%), and North Sumatera (7.55%).

Figure 4.2 presents a brief overview of the Indonesian household budget share composition. The average Indonesian household spends 53.5% of its weekly budget on food, 34% on other goods, 7.5% on transportation, 3.1% on clothing, and

FIGURE 4.2: Weekly Average Household Budget Share




1.9% on fuel. The spending composition is relatively similar with that from Blundell, Duncan, and Pendakur (1998), Leser (1963), and Working (1943). Interestingly, Indonesian households have a larger budget share for food when compared to other communities. Leser (1963) found that the proportion spent on food for Ireland and the U.S. on average is 40.2% and 29.%, while Blundell, Duncan, and Pendakur (1998) found for the U.K. on average is 36.5%. This is presumably because Indonesia has a lower income per-capita. Following Engel law, this implies that most households spend its income on food.

Figure 4.3 shows the household heads' education. Household heads spent approximately 9 years to obtain education. Thirty-five percent of the household heads are only elementary school educated, 28% are senior high educated, 17% are junior high educated, 10% are undergraduate educated, 6% are uneducated, 3% are college diploma educated, 1% are masters educated, and 0.8% are doctorate educated.[1]

Figure 4.5 and 4.4 presents the household heads' sex and marital status. The majority household heads are male (83%). Most of them are also married (81%).

[1]The years of schooling measurement is described in Chapter 3

FIGURE 4.3: Household Head Education (Years)

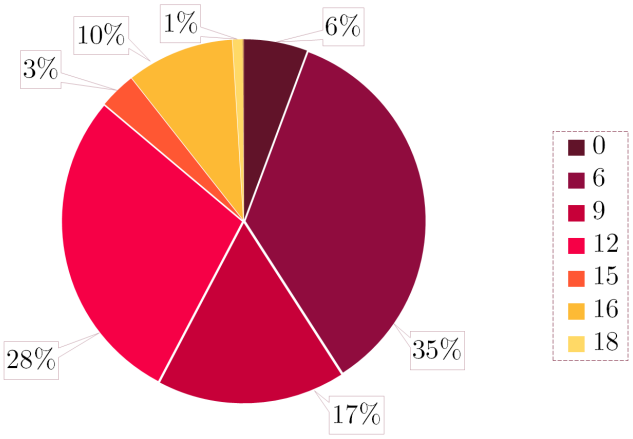



FIGURE 4.4: Household Head Marital Status

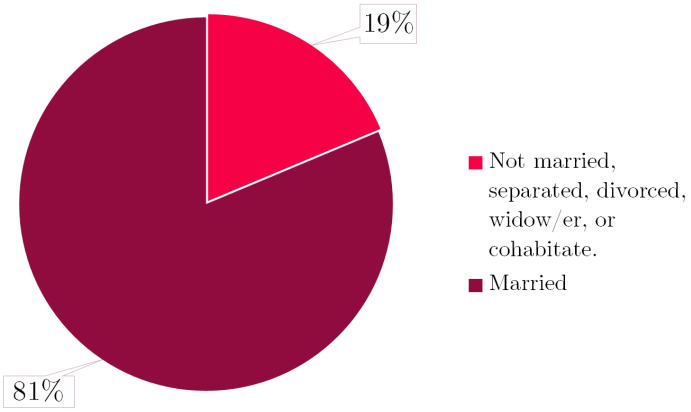
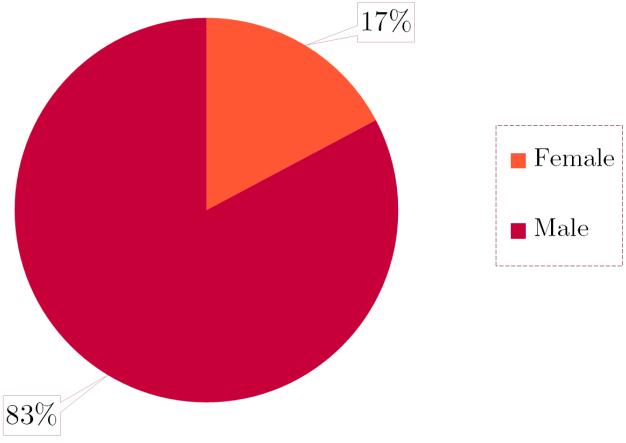



FIGURE 4.5: Household Head Sex



The descriptive statistics of our variables are presented in table 4.2. The maximum share for the selected commodity is 1 or close to 1, while the minimum is 0. The maximum and minimum values are outliers that this study did not remove to show the heterogeneity across household budgets. It may appear unreasonable that some households spend 97% of their budget on clothes and none on food within a week, but this could be because households have already met their basic ne eds through other sources outside of expenditure.

The average weekly household expense is Rp.885,231.51, while the minimum and maximum weekly total expenditure are Rp.1,500 and Rp.30,875,596.15, respectively. The average weekly earned income is Rp.244,624.55, while the minimum and maximum are Rp.10,000 and Rp.25,003,248.75. It may seem unreasonable to have more expenditure than earned income, this is because the fact that the variable earned income does not take into account other sources of income besides labor income. In addition, expenditure might also come from autonomous consumption where households uses their savings.

TABLE 4.2: Descriptive Statistics

| Variable | n | mean | sd | min | max |
|---|---|---|---|---|---|
| Food Share (% expenditure) | 15118 | 0.53 | 0.20 | 0 | 1 |
| Clothing Share (% expenditure) | 15118 | 0.03 | 0.04 | 0 | 0.97 |
| Transport Share (% expenditure) | 15118 | 0.07 | 0.09 | 0 | 0.94 |
| Fuel Share (% expenditure) | 15118 | 0.02 | 0.03 | 0 | 0.86 |
| Others Share (% expenditure) | 15118 | 0.34 | 0.19 | 0 | 1 |
| Weekly Total Expenditure (Rupiah) | 15118 | 885231.55 | 1244104.87 | 1500 | 30875596.15 |
| Weekly Earned Income (Rupiah) | 4428 | 244624.55 | 596157.48 | 10000 | 25003248.75 |
| Household Head Education (years) | 15118 | 9.25 | 4.12 | 0 | 21 |
| Household Head Marital Status (dummy) | 15118 | 0.81 | 0.39 | 0 | 1 |
| Household Head Age (years) | 15118 | 44.67 | 14.51 | 18 | 101 |
| Household Head Sex (dummy) | 15118 | 0.83 | 0.38 | 0 | 1 |
| Household Size (persons) | 15118 | 5.61 | 3.14 | 1 | 25 |

*Note: The number of observations in the study is 15,118 households.*
*Specifically for weekly earned income, the number of observations is 4,428 households.*
*The variable is filtered such that we do not observe non-labor income households*
*and zero labor income households.*
*This is to prevent estimation faulty in the IV estimation.*

The average household head age is 44.67 ≈ 44 years old, while the maximum and minimum are 101 and 18. The household size average is 5.61 ≈ 5 members, while the maximum and minimum are 25 and 1 members.

## 4.2 Estimation Results

### 4.2.1 Baseline Estimation

Table 4.3, 4.4, and 4.5 report the baseline estimates. Table 4.3 is the standard Working-Leser model. We include controls in Table 4.4 and we further include district fixed-effects in Table 4.5. Column (1) reports the estimates when the dependent variable is food share, column (2) when clothing share, column (3) when transportation share, column (4) when fuel share, and (5) when other goods share.

To ensure that the estimation results are best and unbiased, the results reported in all tables have been evaluated to determine whether they meet the Gauss-Markov assumptions. The Gauss-Markov assumption results show the VIF value of the independent and control variables is less than 10, which means the estimation results do not have the perfect multicollinearity problem.

The Breusch–Pagan (BP) test for heteroskedasticity, shows that almost all of the estimates with and without controls have heteroskedastic errors at 5%. After controlling for districts' *fixed-effects*, only food shares have heteroskedastic errors. Nevertheless, the baseline specification models report the corrected standard errors using the *White Robust Standard Errors* to allow for statistical inference under heteroskedastic errors.

Table 4.3 shows that all of the coefficients are statistically significant at 1%. The

TABLE 4.3: Baseline Specification

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | % food | % clothes | % transport | % fuel | % others |
| | (1) | (2) | (3) | (4) | (5) |
| Total expenditure (log) | $-0.078^{***}$ | $-0.004^{***}$ | $0.006^{***}$ | $-0.007^{***}$ | $0.083^{***}$ |
| | (0.002) | (0.001) | (0.001) | (0.0005) | (0.002) |
| Constant | $1.567^{***}$ | $0.079^{***}$ | 0.001 | $0.109^{***}$ | $-0.756^{***}$ |
| | (0.033) | (0.009) | (0.012) | (0.007) | (0.033) |
| Observations | 15,118 | 15,118 | 15,118 | 15,118 | 15,118 |
| $R^2$ | 0.134 | 0.009 | 0.004 | 0.050 | 0.157 |
| Adjusted $R^2$ | 0.134 | 0.009 | 0.003 | 0.050 | 0.157 |
| Residual Std. Error (df = 15116) | 0.184 | 0.035 | 0.087 | 0.028 | 0.178 |
| F Statistic (df = 1; 15116) | $2{,}338.358^{***}$ | $136.837^{***}$ | $53.844^{***}$ | $788.371^{***}$ | $2{,}825.361^{***}$ |

*Note: White Robust Standard Errors are reported in parentheses*
$^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01

signs of food, transportation, fuel and other goods share are consistent with the findings of Leser (1963), namely food share is negative, transportation share is positive, fuel share is negative, and other goods share is positive. In contrast, clothing share has a negative sign and, thus, it is inconsistent with Leser (1963). The R-squared reported on each commodity is low as expected.[2]

In order to reduce omitted variable bias, we include controls in the subsequent estimations and report the results in table 4.4 and 4.5. Both tables show that the signs and significance do not change. Share on food, clothing, and fuel remain negative whereas share on transportation and other foods remain positive.

Table 4.4 shows that the adjusted R-squared improve in all columns. This implies that the controls are necessary to explain the variation in commodity shares. Table 4.5 shows that the adjusted R-squared further increases. This implies that unobserved characteristics that vary across districts are very crucial to understand household consumption pattern.

[2]The model has low explanatory power due to unaccounted heterogeneities. However, the focus of Engel curve estimation is on the relationships between log total expenditure and commodity share.

TABLE 4.4: Baseline Specification with Controls

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | % food | % clothes | % transport | % fuel | % others |
| | (1) | (2) | (3) | (4) | (5) |
| Total expenditure (log) | $-0.074^{***}$ | $-0.005^{***}$ | $0.003^{***}$ | $-0.008^{***}$ | $0.085^{***}$ |
| | (0.003) | (0.001) | (0.001) | (0.001) | (0.003) |
| Urban dummy | 0.004 | $-0.002^{***}$ | $-0.001$ | $0.003^{***}$ | $-0.004$ |
| | (0.003) | (0.001) | (0.002) | (0.0005) | (0.003) |
| Education | $-0.008^{***}$ | $0.001^{***}$ | $0.001^{***}$ | 0.00002 | $0.006^{***}$ |
| | (0.0005) | (0.0001) | (0.0002) | (0.0001) | (0.0004) |
| Marital Status dummy | $-0.002$ | 0.001 | $-0.004$ | $0.007^{***}$ | $-0.003$ |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Age | $-0.001^{***}$ | $-0.0004^{***}$ | $-0.0003^{***}$ | $0.0002^{***}$ | $0.001^{***}$ |
| | (0.0001) | (0.00003) | (0.0001) | (0.00002) | (0.0001) |
| Sex dummy | $0.038^{***}$ | $-0.004^{***}$ | $0.014^{***}$ | $-0.006^{***}$ | $-0.041^{***}$ |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Household Size | $0.004^{***}$ | $0.001^{***}$ | $-0.0004^{*}$ | $0.001^{***}$ | $-0.005^{***}$ |
| | (0.001) | (0.0001) | (0.0002) | (0.0001) | (0.001) |
| Constant | $1.555^{***}$ | $0.113^{***}$ | $0.041^{***}$ | $0.110^{***}$ | $-0.820^{***}$ |
| | (0.034) | (0.010) | (0.013) | (0.007) | (0.034) |
| Observations | 14,896 | 14,896 | 14,896 | 14,896 | 14,896 |
| $R^2$ | 0.166 | 0.042 | 0.013 | 0.087 | 0.190 |
| Adjusted $R^2$ | 0.165 | 0.041 | 0.013 | 0.087 | 0.190 |
| Residual Std. Error (df = 14888) | 0.180 | 0.034 | 0.086 | 0.027 | 0.174 |
| F Statistic (df = 7; 14888) | $421.891^{***}$ | $92.904^{***}$ | $28.468^{***}$ | $203.407^{***}$ | $499.095^{***}$ |

*Note: White Robust Standard Errors are reported in parentheses;*
*Marital Status dummy (1 = Married, 0 = Otherwise); Sex dummy (1 = Male, 0 = Female)*
$^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01

TABLE 4.5: Baseline Specification with Controls and Districts Fixed-Effects

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | % food | % clothes | % transport | % fuel | % others |
| | (1) | (2) | (3) | (4) | (5) |
| Total expenditure (log) | −0.076*** | −0.006*** | 0.003*** | −0.008*** | 0.087*** |
| | (0.003) | (0.001) | (0.001) | (0.001) | (0.003) |
| Urban dummy | 0.002 | −0.002*** | −0.001 | 0.003*** | −0.002 |
| | (0.005) | (0.001) | (0.002) | (0.001) | (0.005) |
| Education | −0.007*** | 0.001*** | 0.001*** | −0.00004 | 0.006*** |
| | (0.0005) | (0.0001) | (0.0002) | (0.0001) | (0.0004) |
| Marital Status dummy | 0.001 | 0.001 | −0.004 | 0.007*** | −0.005 |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Age | −0.0003* | −0.0004*** | −0.0003*** | 0.0002*** | 0.001*** |
| | (0.0001) | (0.00003) | (0.0001) | (0.00002) | (0.0001) |
| Sex dummy | 0.037*** | −0.004*** | 0.013*** | −0.006*** | −0.040*** |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Household Size | 0.003*** | 0.001*** | −0.0002 | 0.0005*** | −0.004*** |
| | (0.001) | (0.0001) | (0.0002) | (0.0001) | (0.001) |
| Constant | 1.315*** | 0.088*** | 0.607*** | 0.108*** | −1.118*** |
| | (0.038) | (0.011) | (0.015) | (0.008) | (0.039) |
| District Fixed-Effects | Yes | Yes | Yes | Yes | Yes |
| Observations | 14,896 | 14,896 | 14,896 | 14,896 | 14,896 |
| $R^2$ | 0.211 | 0.095 | 0.057 | 0.126 | 0.244 |
| Adjusted $R^2$ | 0.195 | 0.076 | 0.037 | 0.108 | 0.228 |
| Residual Std. Error (df = 14592) | 0.177 | 0.034 | 0.085 | 0.027 | 0.170 |
| F Statistic (df = 303; 14592) | 12.873*** | 5.069*** | 2.908*** | 6.941*** | 15.539*** |

*Note: White Robust Standard Errors are reported in parentheses;*
*Marital Status dummy (1 = Married, 0 = Otherwise); Sex dummy (1 = Male, 0 = Female);*
*District Fixed-Effects are dummy variables accounting for 297 cities/districts in 24 observed provinces.*
*p<0.1; **p<0.05; ***p<0.01

The coefficients on household size are significant across models, except for transportation share. The significant positive effect on food and clothing share implies that households with more members present will spend more of their budget on commodities that are not shareable. On the other hand, the significant negative effect on other goods share implies that more members present in the household will be more efficient in allocating their budget for shareable goods. This finding is also consistent with Working (1943), Leser (1963), and Blundell, Duncan, and Pendakur (1998).

### 4.2.2 Quadratic Estimation

Table 4.6 reports the quadratic estimates. The estimation includes all controls and district *fixed-effects*. The estimates do not suffer from perfect multicollinearity. However, with the quadratic term, the errors become heteroskedastic. Thus, the *White Robust Standard Errors* are reported.

With the quadratic form, the interpretation is different. We need to check the sign and significance of log total expenditure and its squared form. The log of expenditure and its squared form are significant in column (1), (3), and (5), whereas they are not in column (2) and (4). This means that the relation on food, transportation, and other goods are non-linear. Food and transportation shares diminish with log total income, while shares of other goods increase with log total income. The quadratic relationship for food is not consistent with the findings of Working (1943), while the linear relationship for clothing is consistent with Blundell, Duncan, and Pendakur (1998).

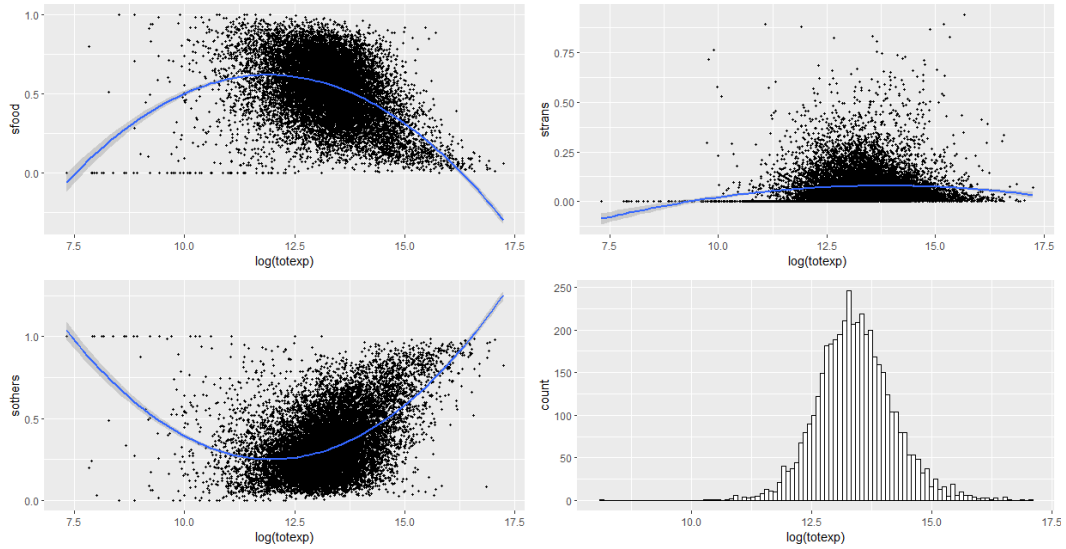
The turning points for food is at 12.35, for transportation is at 12.5, and for other goods is at 11.98 in terms of log weekly total expenditure. Figure 4.6 shows the plotted graphs and the weekly total expenditure distribution. The turning points

TABLE 4.6: Quadratic Form with Controls and District Fixed-Effects

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | % food | % clothes | % transport | % fuel | % others |
| | (1) | (2) | (3) | (4) | (5) |
| Total expenditure (log) | 0.766*** | −0.021 | 0.100*** | −0.006 | −0.839*** |
| | (0.034) | (0.023) | (0.012) | (0.013) | (0.039) |
| Total expenditure (log)^2 | −0.032*** | 0.001 | −0.004*** | −0.0001 | 0.035*** |
| | (0.001) | (0.001) | (0.0005) | (0.0005) | (0.001) |
| Urban dummy | 0.001 | −0.002** | −0.001 | 0.003*** | −0.001 |
| | (0.004) | (0.001) | (0.002) | (0.001) | (0.004) |
| Education | −0.006*** | 0.001*** | 0.001*** | −0.00004 | 0.004*** |
| | (0.0004) | (0.0001) | (0.0002) | (0.0001) | (0.0004) |
| Marital Status dummy | −0.010** | 0.001 | −0.005* | 0.007*** | 0.007 |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Age | 0.0005*** | −0.0004*** | −0.0002*** | 0.0002*** | −0.0001 |
| | (0.0001) | (0.00003) | (0.0001) | (0.00002) | (0.0001) |
| Sex dummy | 0.032*** | −0.004*** | 0.012*** | −0.006*** | −0.034*** |
| | (0.005) | (0.001) | (0.003) | (0.001) | (0.005) |
| Household Size | 0.001*** | 0.001*** | −0.0004 | 0.0005*** | −0.002*** |
| | (0.001) | (0.0001) | (0.0002) | (0.0001) | (0.0005) |
| Constant | −4.215*** | 0.189 | −0.029 | 0.092 | 4.963*** |
| | (0.232) | (0.158) | (0.078) | (0.085) | (0.258) |
| District Fixed-Effects | Yes | Yes | Yes | Yes | Yes |
| Observations | 14,896 | 14,896 | 14,896 | 14,896 | 14,896 |
| $R^2$ | 0.278 | 0.096 | 0.062 | 0.126 | 0.328 |
| Adjusted $R^2$ | 0.263 | 0.077 | 0.042 | 0.108 | 0.314 |
| Residual Std. Error (df = 14591) | 0.169 | 0.034 | 0.085 | 0.027 | 0.160 |
| F Statistic (df = 304; 14591) | 18.520*** | 5.094*** | 3.148*** | 6.919*** | 23.472*** |

*Note: White Robust Standard Errors are reported in parentheses;*
*Marital Status dummy (1 = Married, 0 = Otherwise); Sex dummy (1 = Male, 0 = Female);*
*District Fixed-Effects are dummy variables accounting for 297 cities/districts in 24 observed provinces.*
*p<0.1; **p<0.05; ***p<0.01

FIGURE 4.6: Quadratics Scatter Plot




are well below the mean of households' weekly total expenditure. It is important to note that households with lower expenditure tend to have varying consumption compositions for food and other goods shown by the heteroskedastic errors.

TABLE 4.7: Turning point for Selected Commodities

| | Food | Transportation | Other Goods |
|---|---|---|---|
| Log Total Expenditure* | 12.35 | 12.5 | 11.98 |

### 4.2.3 Instrumental Variable Estimation

Table 4.8 shows the IV estimates. The estimates do not suffer from perfect multicollinearity. The IV estimates have significantly reduced the heteroskedasticity problem where clothing, transportatiom, fuel, and other goods are homoskedastic. Nevertheless, we still report the *White Robust Standard Errors*. The observations are reduced to 4,402 households.

Table B.4 shows the test for weak instruments and the Wu-Hausman endoneity test for the IV estimates. The instrument weekly total earned income $\log y$ is

shown to be a strong instrument for our endogenous variable weekly total expenditure $\log x$ across all selected commodities.

The Wu-Hausman test results shows significance for clothing and transportation share. It implies that weekly total expenditure is endogenous, and therefore, the IV estimates for the two commodities are more consistent than OLS.

On the other hand, The Wu-Hausman test results shows insignificance for food, fuel, and other goods. This implies that although total earned income may have a strong correlation with total expenditure, total expenditure $\log x$ may not be endogenous. In other words, for the three commodities, the variable total earned income might not be a good instrument (see e.g. Battistin and Nadai, 2015).

Nevertheless, the coefficients of the "unbiased" estimates of log total expenditure on clothing and transportation show significance at least at 10%. What differs from previous estimates is that the log total expenditure has a positive effect on clothing share. This implies that, using IV estimates, clothing becomes a normal good.

## 4.3 Discussion

The results of various Engel curve specifications on selected commodities shown to be consistent with Banks et al. (1997), Blundell, Duncan, and Pendakur (1998), De Vreyer et al. (2020), Deaton and Muellbauer (1980b), Leser (1963), and Working (1943).

The findings of heteroskedastic errors in the estimates for Indonesian household data have been found to be consistent with other empirical works on other communities like Allen and Bowley (1935), Lewbel (2008), and Banks et al. (1997).

TABLE 4.8: Instrumental Variable

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | % food | % clothes | % transport | % fuel | % others |
| | (1) | (2) | (3) | (4) | (5) |
| Total expenditure (log) | $-0.100^{***}$ | $0.011^{*}$ | $0.042^{***}$ | $-0.014^{***}$ | $0.061^{**}$ |
| | (0.031) | (0.005) | (0.013) | (0.005) | (0.029) |
| Urban dummy | 0.010 | $-0.006^{***}$ | $-0.008^{*}$ | $0.003^{**}$ | 0.002 |
| | (0.009) | (0.002) | (0.005) | (0.001) | (0.009) |
| Education | $-0.006^{***}$ | $-0.001$ | $-0.001$ | 0.0003 | $0.007^{***}$ |
| | (0.002) | (0.0004) | (0.001) | (0.0003) | (0.002) |
| Marital Status dummy | 0.014 | $-0.006^{**}$ | $-0.024^{***}$ | $0.009^{***}$ | 0.007 |
| | (0.015) | (0.003) | (0.007) | (0.002) | (0.014) |
| Age | $-0.0002$ | $-0.0003^{***}$ | $-0.0002$ | $0.0002^{***}$ | 0.0005 |
| | (0.0003) | (0.0000) | (0.0002) | (0.00005) | (0.0003) |
| Sex dummy | $0.034^{***}$ | $-0.006^{***}$ | $0.010^{*}$ | $-0.002^{*}$ | $-0.035^{***}$ |
| | (0.010) | (0.002) | (0.005) | (0.001) | (0.010) |
| Household Size | $0.004^{***}$ | 0.0003 | $-0.002^{***}$ | $0.001^{***}$ | $-0.003^{**}$ |
| | (0.001) | (0.0002) | (0.001) | (0.0002) | (0.001) |
| Constant | $1.940^{***}$ | $-0.100$ | $-0.446^{**}$ | $0.193^{***}$ | $-0.587^{*}$ |
| | (0.3911) | (0.075) | (0.171) | (0.068) | (0.366) |
| District Fixed-Effects | Yes | Yes | Yes | Yes | Yes |
| Observations | 4,402 | 4,402 | 4,402 | 4,402 | 4,402 |
| $R^2$ | 0.229 | 0.011 | $-0.012$ | 0.178 | 0.260 |
| Adjusted $R^2$ | 0.178 | $-0.054$ | $-0.079$ | 0.124 | 0.212 |
| Residual Std. Error (df = 4130) | 0.168 | 0.035 | 0.087 | 0.020 | 0.163 |

*Note: White Robust Standard Errors are reported in parentheses;*
*Log earned income is used as the instrument for log total expenditure;*
*Marital Status dummy (1 = Married, 0 = Otherwise); Sex dummy (1 = Male, 0 = Female);*
*District Fixed-Effects are dummy variables accounting for 297 cities/districts in 24 observed provinces;*
*The number of observations drops to 4,402 because we only include non-zero HH incomes;*
*R-Squared is meaningless in IV estimates.*
$^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01

The calculated income elasticity of demand for each specification is presented in Table 4.9. In this empirical Engel curve estimation, it is concluded that among empirical specifications, food is considered as a necessity, which concurs with the theory of Engel law (Zimmerman, 1932). Further, transportation and other goods is considered a luxury good, while fuel is a necessity. The calculated income elasticity of clothing for the first three models was inelastic (necessity), but with the IV estimation, it is elastic (luxury good).

TABLE 4.9: Calculated Average Income Elasticity of Demand

| | Food | Clothing | Transportation | Fuel | Other Goods |
|---|---|---|---|---|---|
| Baseline | 0.854 | 0.884 | 1.074 | 0.642 | 1.242 |
| Baseline (Controls) | 0.861 | 0.830 | 1.035 | 0.583 | 1.247 |
| Baseline (Controls + Fixed Effects) | 0.856 | 0.816 | 1.040 | 0.578 | 1.255 |
| Instrumental Variable | 0.810 | 1.354 | 1.531 | 0.212 | 1.178 |

*Note: Quadratics model is not included because the estimates on clothing and fuel share are not significant*
*Elasticities description: Inferior* $\eta_i < 0$*, Necessity* $0 < \eta_i < 1$*, and Luxury* $\eta_i > 1$

# CHAPTER V

# CONCLUSION

## 5.1 Conclusion

The empirical evidence of consumption patterns in Indonesian households across 24 provinces shows that the estimation results show similarities with the household budget in other communities such as Banks et al. (1997), Blundell, Duncan, and Pendakur (1998), De Vreyer et al. (2020), Deaton and Muellbauer (1980b), Leser (1963), and Working (1943). It is shown that food is a necessity, proving the Engel law. Clothing and fuel are also necessities, while transportation and other goods are luxury goods. The shape of the Engel curve for food, transportation, and other goods is non-linear, while clothing and fuel are linear.

The effect on household size for shareable goods contained in the "other goods" share was found to be negative. The larger the family size, the more efficient it is for households to purchase things like furniture, vehicles, housing maintenance, electricity, water, etc. In contrast, non-shareable goods, like food, are positive. The larger the family size, the more is spent on food.

The different empirical specifications used for Indonesian households are shown to fit in almost all models. Specifically, the use of instrumental variables for clothing and transportation are shown to help with the bias. The calculated income elasticity of demand for clothing and transportation is corrected, where clothing are considered as luxury goods in the IV estimation.

Consistent with Lewbel (2008), in the case of Indonesia, the estimates on Engel

curves are found to be heteroskedastic and endogenous. Heterogeneity in taste and prices across communities is still present even with the inclusion of controls and district *fixed-effects*. The variations in errors and bias might be due to the zero expenditure problem.

## 5.2 Limitations

The limitations of this study include:

1. prices and quality of goods are not accounted for in the Engel curve analysis, resulting in the estimates being heteroskedastic;
2. the study only covers cross-section data in the years 2014−2015;
3. households are not divided into income groups to see the differences in consumption patterns and income elasticities across household income;
4. the instrumental variable weekly total earned income does not include other sources of income aside from wage, salary, or income, thus reducing the observations down to 4,402 households. The variable might not be the best instrument because the variable in it self might be correlated with the structural model's error term.

## 5.3 Suggestion for Further Research

Based on the limitations of this study, the suggestions for further research are as follows:

1. the inclusion of prices of commodities, and thus the use of demand systems using Deaton's Almost Ideal Demand System (AIDS) due to Deaton and

Muellbauer (1980a) allowing for both income and price elasticities among commodities;

2. the use of panel data across IFLS waves for welfare analysis;
3. dividing the observations into income groups;
4. improving the instrumental variable with the use of other sources of income.

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# APPENDIX A

## Engel Aggregation

Suppose a utility function for goods $x_i$

$$U(x_1, x_2, \ldots, x_i) \tag{A.1}$$

To proof for the Engel Aggregation, suppose we have a consumer budget of bundles of goods $i$ where $m$ is defined as income (total expenditure), while $p_i$ and $x_i$ is defined the price and quantity purchased of good $i$.

$$g(x) = m = p_1x_1 + p_2x_2 + \cdots + p_ix_i \tag{A.2}$$

$$m = \sum_{i=1}^{N} p_ix_i \tag{A.3}$$

For Marshallian demand function, we assume that consumers maximize their utility function subject to their budget constraint.

$$\max\ U(x)\ s.t.\ g(x) \tag{A.4}$$

Solving (A.4) using the method Lagrange multipliers, we obtain a Marshallian demand function for goods $i$,

$$\mathscr{L} = U(x_1, x_2, \ldots, x_i) - \lambda[(\sum_{i=1}^{N} p_ix_i) - m] \tag{A.5}$$

To find the optimum values for $x_i$'s

$$\nabla U(x_1, x_2, \ldots, x_i) = C_n = (\frac{\partial U}{\partial x_1}, \frac{\partial U}{\partial x_2}, \ldots, \frac{\partial U}{\partial x_i}) \tag{A.6}$$

$$\nabla g(x_1, x_2, \ldots, x_i) = C = (\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \ldots, \frac{\partial g}{\partial x_i}) \tag{A.7}$$

Optimum, when $\nabla U(x_1, x_2, \ldots, x_i) = \lambda \nabla g(x_1, x_2, \ldots, x_i)$. In order to obtain the optimum, we taking the first order conditions for $\mathscr{L}$ with respect to $x_i$'s and $\lambda$

$$\begin{aligned} \frac{\partial \mathscr{L}}{\partial x_1} &= \frac{\partial U_{x_1}}{\partial x_1} - \lambda p_1 = 0 \\ \frac{\partial \mathscr{L}}{\partial x_2} &= \frac{\partial U_{x_2}}{\partial x_2} - \lambda p_2 = 0 \\ &\vdots \\ \frac{\partial \mathscr{L}}{\partial x_i} &= \frac{\partial U_{x_i}}{\partial x_i} - \lambda p_i = 0 \end{aligned} \tag{A.8}$$

$$\frac{\partial \mathscr{L}}{\partial \lambda} = \sum_{i=1}^{N} p_i x_i - m = 0 \tag{A.9}$$

Equation (A.8) can also be written as

$$\lambda = \frac{MU_{x_1}}{p_1} = \frac{MU_{x_2}}{p_2} = \cdots = \frac{MU_{x_i}}{p_i} \tag{A.10}$$

Given a utility function and solving $\lambda$ for (A.8) into our budget constraint, $x_i$ will be at its optimum values, in which in itself is a function of $p_1, p_2, \ldots, p_i, m$ and result in a series of Marshallian demand functions

$$\begin{aligned} x_1^* &= x_1(p_1, p_2, \ldots, p_i, m) \\ x_2^* &= x_2(p_1, p_2, \ldots, p_i, m) \\ &\vdots \\ x_i^* &= x_1(p_1, p_2, \ldots, p_i, m) \end{aligned} \tag{A.11}$$

For simplicity of exposition, suppose our Marshallian demand function is given as

$$x_i^* = x_i(p_i, m) \tag{A.12}$$

Substituting our demand function (A.12) into our budget constraint $g(x)$,

$$m = p_1 x_1(p_1, m) + p_2 x_2(p_2, m) + ... + p_i x_i(p_i, m) \tag{A.13}$$

Differentiating both side with respect to $m$.

$$1 = p_1 \frac{\partial x_1}{\partial m} + ... + p_i \frac{\partial x_i}{\partial m} \tag{A.14}$$

$$1 = \sum_{i=1}^{N} p_i \frac{\partial x_i}{\partial m} \tag{A.15}$$

Multiply each good $i$ by $\frac{x_i m}{x_i m}$, we get

$$1 = \sum_{i=1}^{N} p_i \frac{\partial x_i}{\partial m} \frac{x_i m}{x_i m} \tag{A.16}$$

$$1 = \sum_{i=1}^{N} \frac{p_i x_i}{m} \frac{\partial x_i}{\partial m} \frac{m}{x_i}, \tag{A.17}$$

where, the expenditure share on good $i$ is $w_i = \frac{p_i x_i}{m}$ and the income elasticity of demand for good $i$ is given by $\eta_i = \frac{\partial x_i}{\partial m} \frac{m}{x_i}$

Re-writing (A.17),

$$\sum_{i=1}^{N} w_i \eta_i = 1 \tag{A.18}$$

## Calculating Income Share Elasticity of Demand

Suppose a Working-Leser Engel curve specification for good $i$, where $w_i$ is expenditure share on good $i$ and $m$ is total expenditure:

$$w_i = \alpha_i + \beta_i \log m + \varepsilon_i \tag{A.19}$$

Take the derivative of $\log(w)$ with respect to $\log(m)$ to calculate the income share elasticity

$$\log(w) = \log(\alpha + \beta_i \log(m)) \tag{A.20}$$

$$\frac{\partial \log(w)}{\partial \log(m)} = \frac{\partial(\log(\alpha + \beta_i \log(m))}{\partial \log(m)} \tag{A.21}$$

$$\eta_{w_i} = \frac{\beta_i}{(\alpha + \beta_i \log(m))} \tag{A.22}$$

$$\eta_{w_i} = \frac{\beta_i}{w} \tag{A.23}$$

## Deriving Income Elasticity of Demand

Let $w_i$ denote the budget share of good $i$ defined as $w_i = \dfrac{p_i x_i}{m}$. Where $p_i$ and $x_i$ denote as prices and quantity of goods $i$. While $m$ denoted as the total expenditure, which is defined as:

$$m = \sum_{i=1}^{n} p_i x_i \tag{A.24}$$

The income elasticity of demand of good $i$ is defined as:

$$\eta_i = \frac{\partial x_i}{\partial m} \frac{m}{x_i}, \tag{A.25}$$

where $\dfrac{x_i}{m} = \dfrac{w_i m}{p_i}$ and $\dfrac{m}{x_i} = \dfrac{p_i}{w_i}$.

Rewriting the Equation (A.25)

$$\eta_i = \frac{\partial(\frac{w_i m}{p_i})}{\partial p_i}\frac{p_i}{w_i} \tag{A.26}$$

Using the quotient and power rule

$$\eta_i = \left[\frac{\frac{\partial(w_i m)}{\partial m}p_i - w_i m\frac{\partial p_i}{m}}{p_i^2}\right]\frac{p_i}{w_i} \tag{A.27}$$

$$= \left[\frac{\left[\frac{\partial w_i}{\partial m}m + w_i\right]p_i - w_i m\frac{\partial p_i}{\partial m}}{p_i^2}\right]\frac{p_i}{w_i} \tag{A.28}$$

$$= \left[\left[\frac{\partial w_i}{\partial m}m + w_i\right]\frac{1}{p_i} - w_i m\frac{\partial p_i}{\partial m}\frac{1}{p_i^2}\right]\frac{p_i}{w_i} \tag{A.29}$$

$$= \left[\frac{\partial w_i}{\partial m}\frac{m}{p_i} + \frac{w_i}{p_i} - \frac{w_i m}{p_i^2}\frac{\partial p_i}{\partial m}\right]\frac{p_i}{w_i} \tag{A.30}$$

$$= \left[\partial w_i \frac{m}{\partial m}\frac{1}{p_i} + \frac{w_i}{p_i} - \frac{w_i}{p_i^2}\partial p_i \frac{\partial m}{m}\right]\frac{p_i}{w_i} \tag{A.31}$$

$$= \frac{\partial w_i}{w_i}\frac{m}{\partial m} + 1 - \frac{\partial p_i}{p_i}\frac{m}{\partial m} \tag{A.32}$$

Assuming that unit value does not change, $\frac{\partial p_i}{p_i} = 0$. And the income share elasticity of demand $\eta_{w_i} = \frac{\partial w_i}{\partial m}\frac{m}{w_i} = \frac{\beta_i}{w_i}$

$$\begin{aligned} \eta_i &= \frac{\partial w_i}{\partial m}\frac{m}{w_i} + 1 \\ \eta_i &= \eta_{w_i} + 1 \end{aligned} \tag{A.33}$$

Thus, we have the elasticity of good $i$ defined as (Teklu, Johnson, et al., 1986)

$$\eta_i = \frac{\beta_i}{\bar{w}_i} + 1 \tag{A.34}$$

The following are the classification of goods:

- Inferior: $\eta_i < 0$;
- Necessities: $0 < \eta_i < 1$;
- Luxury: $\beta_i > 1$.

# APPENDIX B

## Multicollinearity

TABLE B.1: Multicollinearity Test

| Dependent variables | GVIF | df | GVIF$\hat{}$(1/(2df)) |
|---|---|---|---|
| log(totexp) | 23.38 | 1 | 4.83 |
| urban | 3.06 | 1 | 1.75 |
| HHeduc | 7.89 | 1 | 2.80 |
| HHstatus | 3.93 | 1 | 1.98 |
| HHage | 3.12 | 1 | 1.76 |
| HHsex | 1.69 | 1 | 1.30 |
| HHsize | 2.76 | 1 | 1.66 |
| factor(kabs) | 45.19 | 264 | 1.00 |

*Note: GVIF (Generalized Variance Inflation Factor) is the square root of the VIF*

TABLE B.2: Correlation Matrix

| | sfood | sclothes | strans | sfuel | sothers | totexp | income | HHeduc | HHstatus | HHage | HHsex | HHsize |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| sfood | 1.00 | | | | | | | | | | | |
| sclothes | -0.11 | 1.00 | | | | | | | | | | |
| strans | -0.29 | 0.01 | 1.00 | | | | | | | | | |
| sfuel | 0.02 | -0.01 | -0.04 | 1.00 | | | | | | | | |
| sothers | -0.88 | -0.08 | -0.15 | -0.15 | 1.00 | | | | | | | |
| totexp | -0.40 | -0.08 | -0.01 | -0.13 | 0.44 | 1.00 | | | | | | |
| income | -0.02 | 0.00 | 0.02 | -0.01 | 0.01 | 0.04 | 1.00 | | | | | |
| HHeduc | -0.27 | 0.06 | 0.08 | -0.14 | 0.25 | 0.25 | 0.02 | 1.00 | | | | |
| HHstatus | -0.05 | -0.02 | 0.03 | -0.03 | 0.04 | 0.12 | 0.01 | 0.07 | 1.00 | | | |
| HHage | 0.08 | -0.14 | -0.08 | 0.16 | -0.04 | -0.03 | -0.03 | -0.41 | -0.11 | 1.00 | | |
| HHsex | -0.02 | -0.04 | 0.07 | -0.09 | 0.01 | 0.10 | 0.03 | 0.14 | 0.60 | -0.12 | 1.00 | |
| HHsize | 0.03 | -0.01 | -0.04 | 0.07 | -0.02 | 0.07 | 0.02 | -0.18 | 0.07 | 0.39 | 0.00 | 1.00 |

## Heteroskedasticity Test

TABLE B.3: Studentized Breusch-Pagan Test

| | BP | df | p-value | Heteroskedastic |
|---|---|---|---|---|
| Baseline Model | | | | |
| *Food share* | 436.99 | 1 | 2.20E-16 | Yes |
| *Clothing share* | 100.12 | 1 | 2.20E-16 | Yes |
| *Transportation share* | 0.74 | 1 | 0.3886 | No |
| *Fuel share* | 89.29 | 1 | 2.20E-16 | Yes |
| *Other goods share* | 35.27 | 1 | 2.86E-09 | Yes |
| Baseline Model with Controls | | | | |
| *Food share* | 501.96 | 7 | 2.20E-16 | Yes |
| *Clothing share* | 108.19 | 7 | 2.20E-16 | Yes |
| *Transportation share* | 31.964 | 7 | 4.12E-05 | Yes |
| *Fuel share* | 151.02 | 7 | 2.20E-16 | Yes |
| *Other goods share* | 204.63 | 7 | 2.20E-16 | Yes |
| Baseline Model with Controls and District Fixed Effects | | | | |
| *Food share* | 867.76 | 303 | 2.20E-16 | Yes |
| *Clothing share* | 320.74 | 303 | 0.2315 | No |
| *Transportation share* | 337.72 | 303 | 0.08275 | No |
| *Fuel share* | 299.57 | 303 | 0.545 | No |
| *Other goods share* | 499.23 | 303 | 8.54E-12 | Yes |
| Quadratics Model | | | | |
| *Food share* | 857.87 | 304 | 2.20E-16 | Yes |
| *Clothing share* | 791.39 | 304 | 2.20E-16 | Yes |
| *Transportation share* | 350.83 | 304 | 0.03317 | Yes |
| *Fuel share* | 519.32 | 304 | 1.75E-13 | Yes |
| *Other goods share* | 890.04 | 304 | 2.20E-16 | Yes |
| Instrumental Variables Model | | | | |
| *Food share* | 312.59 | 271 | 0.0417 | Yes |
| *Clothing share* | 261.08 | 271 | 0.6564 | No |
| *Transportation share* | 261 | 271 | 0.6578 | No |
| *Fuel share* | 195.94 | 271 | 0.9998 | No |
| *Other goods share* | 308.28 | 271 | 0.0591 | No |

## Diagnostic Tests

TABLE B.4: Diagnostic Test for Instrumental Variables

| | statistic | p-value |
|---|---|---|
| Food Share | | |
| Weak Instruments | 79.193 | <2e-16*** |
| Wu-Hausman | 0.433 | 0.51 |
| Clothing Share | | |
| Weak Instruments | 79.193 | <2e-16*** |
| Wu-Hausman | 9.069 | 0.00262** |
| Transportation Share | | |
| Weak Instruments | 79.19 | <2e-16 *** |
| Wu-Hausman | 10.43 | 0.00125 ** |
| Fuel Share | | |
| Weak Instruments | 79.19 | <2e-16 *** |
| Wu-Hausman | 1.60 | 0.206 |
| Other Goods Share | | |
| Weak Instruments | 79.193 | <2e-16 *** |
| Wu-Hausman | 1.971 | 0.16 |

*Note: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$*

# APPENDIX C

## Scatter Plots

FIGURE C.1: Engel Curves Scatter Plots for Selected Commodities

